

Part 4: C Code Implementation Deep Dive

Mathematical Physics, SocketCAN Multiplexing & Full Protocol Matrix

1. The 10-Node CAN Protocol Matrix (Standard 8-Byte Packed Frames)

All messages adhere to standard 11-bit identifiers and strict ``#pragma pack(push, 1)`` structs defined in ``common/vehicle_protocol.h``:

CAN ID	Message Name	Bus	Signals & Fixed-Point Encoding
0x100	PcmTelemetryMsg	vcan0	Speed (kph*10), Motor RPM, Actual Torque (Nm), Motor Temp (°C), Gear.
0x101	PcmCmdMsg	vcan0	Target Torque (Nm), Speed Limit, Inverter Enable, Regen Level.
0x110	BmsTelemetryMsg	vcan0	SoC %, Pack Voltage (V*10), Pack Current (A*10), Max Cell Temp, Status, SoH.
0x120	BrakeTelemetryMsg	vcan0	Brake Torque (Nm), ABS Active Mask (FL/FR/RL/RR), Pressure (bar), Rotor Temp (°C).
0x121	BrakeCmdMsg	vcan0	Req Brake Torque (Nm), Emergency Brake Enable, Parking Brake Request.
0x130	SteerTelemetryMsg	vcan0	Actual Pinion Angle (deg), Motor Assist (Nm*10), Hand Torque (Nm*10), LKA State.
0x131	SteerCmdMsg	vcan0	Driver Steering Angle (deg), LKA Corrective Torque Overlay, Steering Mode.
0x300	AdasTelemetryMsg	vcan0	Lead Distance (m*10), Rel Speed (kph*10), TTC (s*10), FCW Alert, AEB Request, LDW.
0x301	AdasCmdMsg	vcan0	ADAS Mode, Emergency Braking Ack, Lane Keeping Assist Enable.
0x200	BcmTelemetryMsg	vcan1	Doors Locked bitmask, Lights Active bitmask, Cabin Temp, Ambient Lux, Wipers.
0x201	BcmCmdMsg	vcan1	Lock Command (Lock/Unlock), Light Command (LowBeam/Hazard/Brake), Horn.
0x210	HmiInputMsg	vcan1	Throttle %, Brake %, Steering Angle, Selected Gear (P/R/N/D), Drive Mode.
0x220	HvacTelemetryMsg	vcan1	Cabin Temp, Evaporator Temp, Coolant Loop Temp, Compressor Power (W), Blower RPM.
0x221	HvacCmdMsg	vcan1	Target Cabin Temp, Fan Speed (0-7), AC Compress Enable, Battery Chill Req.
0x400	TelematicsStatusMsg	vcan1	5G CSQ (0-31), Cloud Connect State, OTA State, OTA Progress %, GNSS Fix, Ping ms.
0x401	TelematicsCmdMsg	vcan1	Ack Command, Firmware Version, Diagnostic DTC Count, Sync Rate.

2. Dual-Homed POSIX select() Multiplexing

In ``nodes/central_compute.c``, the controller reads concurrently from both domain buses without blocking:

```

fd_set read_fds;
FD_ZERO(&read_fds);
FD_SET(sock_pt, &read_fds); // vcan0: Powertrain, Chassis & Safety
FD_SET(sock_body, &read_fds); // vcan1: Body, Cockpit, HVAC & Telematics

struct timeval tv = { .tv_sec = 0, .tv_usec = 10000 }; // 10ms timeout
int ret = select(max_fd + 1, &read_fds, NULL, NULL, &tv);
if (ret > 0) {
    if (FD_ISSET(sock_pt, &read_fds)) {
        // Read 0x100, 0x110, 0x120, 0x130, 0x300 frames
    }
    if (FD_ISSET(sock_body, &read_fds)) {
        // Read 0x200, 0x210, 0x220, 0x400 frames
    }
}
}

```

3. Mathematical Physics & Safety Algorithms

- **Radar Time-To-Collision (TTC):** In `adas_node.c``, Doppler closing velocity calculates $TTC = \frac{d_{\text{lead}}}{v_{\text{closing}}}$. If $TTC < 1.3 \text{ s}$, AEB Full Emergency is asserted.
- **Braking Blending & ABS Slip:** In `brake_node.c``, master cylinder pressure translates to torque ($\tau = P \cdot 21.5$). At high deceleration, wheel slip $s = \frac{v_{\text{veh}} - v_{\text{wheel}}}{v_{\text{veh}}} > 0.18$ activates 15Hz solenoid valve dumping.
- **EPS Speed-Sensitive Assist:** Assist gain scales dynamically with vehicle velocity: $K_{\text{assist}}(v) = 3.6 - 2.2 \cdot \frac{\min(v, 120)}{120}$, delivering light parking effort and stiff highway stability.
- **Battery Coulomb Counting & Joule Heating:** In `bms_node.c``, power is converted into current $I_{\text{pack}} = \frac{P_{\text{elec}}}{V_{\text{pack}}}$, with thermal heating $P_{\text{heat}} = I_{\text{pack}}^2 \cdot R_{\text{int}}$.
- **OTA State Machine:** In `telematics_node.c``, firmware updates cycle from IDLE -> DOWNLOADING -> VERIFYING -> FLASHING (A/B dual partition) -> COMPLETE.